HEALTHCARE DATA ANALYTICS REPORT

Factors Associated with Hospital Length of Stay Using Python

A Healthcare Data Analytics Project: Data Cleaning, Hypothesis Testing, Correlation, Regression and Effect-Size Analysis

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Dataset: Synthetic sample of 50,000 hospital admission records (48,019 after cleaning) | Tools: Python (pandas, NumPy, SciPy, Matplotlib)

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Executive Summary

Length of stay (LOS) is one of the most important indicators of hospital efficiency. Long stays increase the cost of care, keep beds occupied that other patients are waiting for, expose patients to hospital-acquired infections, and add to the workload of nurses and doctors. For a busy Nigerian teaching hospital, understanding which patients tend to stay longer makes it easier to plan beds, staff and discharge services.

This project analysed a synthetic dataset of 50,000 hospital admission records, created with Claude AI to resemble the records of a Nigerian hospital, to identify the factors associated with length of stay. The file was deliberately created with real-world data problems (inconsistent spellings, impossible values, missing values, mixed date formats and duplicates). After a documented cleaning process, 48,019 records remained. Using Python, the project applied descriptive statistics, t-tests, the Mann–Whitney U test, ANOVA, the Kruskal–Wallis test, Pearson and Spearman correlation, effect sizes and multiple linear regression on log-transformed length of stay with a train–test evaluation.

The average length of stay was 5.50 days (median 4 days, range 1–46 days). Primary diagnosis, comorbidity count, age, ward, previous admissions, admission type and 30-day readmission were significantly associated with length of stay, while gender and payment type were not. Diagnosis was the strongest factor: tuberculosis (7.98 days), sepsis (7.43 days) and stroke (7.27 days) had the longest stays, against 4.02 days for gastroenteritis. ICU patients stayed about 2.4 days longer than medical-ward patients. The regression model explained about 19.7% of the variation in log-transformed length of stay and performed the same on unseen test data ($R^2 = 0.199$), so most of the variation comes from factors not recorded in this dataset. The data are synthetic and the results show associations, not causes, so the findings are an analytical demonstration rather than clinical evidence.

Key Findings at a Glance

48,019	5.50 days	Diagnosis	19.7%
Admissions Analysed	Mean Length of Stay (median 4 days)	Strongest Factor Associated with LOS	Variation in log-LOS Explained by the Model

Factor	Significant?	Effect on Length of Stay	Recommended Action
Primary Diagnosis	Yes ($p < 0.001$)	Tuberculosis 7.98 days vs gastroenteritis 4.02 days; $\eta^2 = 0.081$ (medium)	Diagnosis-specific discharge pathways
Comorbidity Count	Yes ($p < 0.001$)	Rises from 4.34 days (none) to 7.81 days (5+); $\rho = 0.20$	Early multidisciplinary review
Age	Yes ($p < 0.001$)	Rises from 4.37 days (5–17 years) to 6.93 days (65+); $\rho = 0.19$	Early discharge planning for older patients
Ward	Yes ($p < 0.001$)	ICU 7.74 days vs 5.32 days on medical wards	Review ICU step-down criteria
Previous Admissions	Yes ($p < 0.001$)	+0.39 days per extra admission; $\rho = 0.08$	Case management for frequent admitters
Admission Type	Yes ($p < 0.001$)	Emergency 5.85 days vs elective 5.08 days	Streamline emergency flow
30-Day Readmission	Yes ($p < 0.001$)	Readmitted patients stayed 2.05 days longer ($d = 0.51$, medium)	Check discharge readiness
Gender	No ($p = 0.595$)	Almost identical (+0.02 days)	No targeted action needed
Payment Type	No ($p = 0.441$)	Almost identical across NHIS, HMO and self-pay	No targeted action needed

Full details and additional recommendations are provided in the Results, Limitations, Recommendations and Next Steps sections of this report.

Business Problem

Hospitals aim to manage length of stay carefully because long and unpredictable stays:

- Increase the cost of care and the use of hospital resources.
- Keep beds occupied, which delays admissions from the emergency unit and the waiting list.
- Increase the workload of nurses, doctors and support staff.
- May point to delays in diagnosis, treatment, discharge planning or care coordination.

The hospital wants to identify the patient and admission factors associated with length of stay so that management can predict which patients are likely to need longer stays, plan beds and staffing more effectively, and focus discharge planning on the patients who need it most.

Project Objective

To identify which patient and admission characteristics (age, gender, primary diagnosis, number of previous admissions, 30-day readmission status, admission type, ward, comorbidity count and payment type) are associated with length of stay, to measure how large each association is, and to evaluate how much of the variation in length of stay these factors explain together.

Dataset and Data Source

The dataset used in this project is synthetic. It was generated with Claude AI (Anthropic) from a written prompt describing the variables listed below, so it contains no real patient information. The raw file has 50,000 rows (one row per admission record) and 12 columns. It was deliberately built to be "dirty" in the ways that real hospital records often are, so that the data-cleaning process could be demonstrated. After cleaning, 48,019 records remained. The outcome variable is length_of_stay; the other 11 variables are potential factors.

Variable	Type	Description
patient_id	Identifier	Unique patient code (PT000001 …), used only to detect duplicates
age	Numerical	Age in years (0–105 after cleaning)
gender	Categorical	Female, Male, Unknown
diagnosis	Categorical	14 primary diagnoses plus "Unknown" (e.g. malaria, typhoid fever, stroke, tuberculosis)
previous_admissions	Numerical	Number of previous hospital admissions
readmitted_30d	Categorical	Readmitted within 30 days of discharge (Yes / No / Unknown)
length_of_stay	Outcome	Number of days spent in hospital (1–180 allowed)
admission_type	Categorical	Emergency, Elective, Referral, Unknown
ward	Categorical	Medical, Surgical, Paediatric, Obstetrics & Gynaecology, ICU, Emergency, Unknown
comorbidity_count	Numerical	Number of coexisting medical conditions
payment_type	Categorical	NHIS, Out-of-pocket, HMO, Waiver/Exempt, Unknown
admission_date	Date	Date of admission (2023–2025)

Because the records are synthetic, the findings in this report describe patterns within this dataset and are not clinical evidence.

Data Cleaning and Documentation

Every cleaning decision was recorded so that the process is transparent and repeatable. The raw file contained 50,000 rows and 12 columns. The table below lists each problem found, the action taken and the number of records affected. Counts for each problem are calculated at the point in the workflow where that step was applied, so they should not be added together.

Cleaning Principles

- **Never guess the outcome.** Records with a missing or impossible length of stay were removed, not imputed.
- **Fix what can be fixed safely.** Spelling, capitalisation, abbreviations and formats were standardised using documented mapping rules.
- **Do not invent categories.** Missing or vague categorical values were labelled "Unknown" instead of being filled in.
- **Use the median for skewed numbers.** Missing age, previous admissions and comorbidity counts were filled with the median of the valid values.
- **Impossible is not the same as rare.** Values were removed only when they were clinically impossible (negative ages, 999 days). Long but plausible stays were kept.

Cleaning Log

Step	Problem found	Action taken	Records
1	Leading/trailing spaces in text fields	Stripped whitespace from all columns	2,752
2	Exact duplicate records (all 12 fields identical)	Removed duplicates, kept first occurrence	516
2	Repeated patient_id with minor formatting differences	Removed repeats, kept first occurrence	234
3	Inconsistent gender labels (M, f, MALE, female ...)	Standardised to Male / Female	4,588
3	Placeholder (Unknown, ?, N/A) or missing gender	Labelled "Unknown"	992
4	Ages stored as text ("45 yrs", "45y")	Extracted the numeric value	684
4	Impossible ages (negative, 150, 200, 999)	Set to missing (valid range 0-105)	238
4	Missing age (original blanks + invalid values)	Median imputation (40 years)	957
5	Inconsistent diagnosis names (case, spelling, abbreviations: HTN, CCF, RTA, DM)	Mapped to 14 standard diagnosis names	12,168
5	Vague entries ("Not stated", "Others", "??") or missing diagnosis	Labelled "Unknown"	1,229
6	Counts written as words ("none", "once", "two")	Converted to numbers	243
6	Impossible previous-admission counts (-1, -2, 50, 99)	Set to missing (valid range 0-30)	197
6	Missing previous admissions	Median imputation (1)	1,667
7	Inconsistent readmission coding (Y, yes, 1, True / N, no, 0, False)	Standardised to Yes / No	7,167
7	Missing readmission status	Labelled "Unknown" (excluded from the Yes/No comparison)	1,481
8	Length of stay stored as text ("3 days", "3d")	Extracted the numeric value	1,926
8	Fractional length of stay (x.5 days)	Rounded to whole days (half-days rounded up)	1,439
8	Impossible length of stay (0, negative, 365, 500, 999)	Set to missing (valid range 1-180 days)	286
8	Missing length of stay (original blanks + invalid values)	Rows removed (the outcome variable is never imputed)	1,231
9	Inconsistent admission type labels (EMERGENCY, Emerg., Refered)	Standardised to Emergency / Elective / Referral	4,599
9	Missing admission type	Labelled "Unknown"	1,907
10	Inconsistent ward names (Med Ward, Surg, O&G, A&E, icu)	Standardised to 6 wards	5,281
10	Missing ward	Labelled "Unknown"	1,440
11	Impossible comorbidity counts (-1, 25, 99)	Set to missing (valid range 0-15)	195
11	Missing comorbidity count	Median imputation (2)	2,585
12	Inconsistent payment labels (nhis, N.H.I.S, OOP, Cash, Waiver)	Standardised to NHIS / Out-of-pocket / HMO / Waiver-Exempt	5,550
12	Missing payment type	Labelled "Unknown"	1,912
13	Mixed date formats (dd/mm/yyyy, dd-Mon-yyyy, mm/dd/yy, yyyy/mm/dd)	Parsed and converted to ISO format (YYYY-MM-DD)	7,046
13	Invalid or future dates ("0000-00-00", "unknown", 31/02/2024, 2030)	Set to missing (left blank; date is not used as a predictor)	145
13	Missing admission date	Left blank (not imputed)	1,104

Before and After Cleaning

Measure	Raw data	Clean data
Rows	50,000	48,019
Columns	12	13 (age_group added)
Duplicate records	750	0
Rows removed for missing or impossible length of stay	—	1,231
Age values stored as text or impossible	Yes	None; all numeric, 0–98 years
Distinct diagnosis spellings	72 variants	14 standard names + Unknown
Date formats	5 formats mixed	One format (YYYY-MM-DD)
Missing values in analysis variables	Present in 10 of 12 columns	None (except 1,104 blank admission dates)

In total, 750 duplicate records and 1,231 records with no usable length of stay were removed, leaving 48,019 records (96.0% of the original file). A new variable, age_group (0–4, 5–17, 18–34, 35–49, 50–64, 65+ years), was created from age. The cleaned dataset and the cleaning log are provided with this report.

Methodology

This project followed a structured healthcare data analytics workflow.

1. Data Preparation

The raw CSV file was imported into Python and inspected for structure, data types, missing values and duplicates. The cleaning steps documented above were then applied in a fixed order. The final dataset has 48,019 rows and 13 columns.

2. Descriptive Statistics

Frequencies, percentages, means, medians, standard deviations, quartiles, skewness and kurtosis were calculated. Length of stay is right-skewed, so medians are reported next to means and non-parametric tests are used alongside parametric tests.

3. Hypothesis Testing

For every factor the null hypothesis was that the factor is not associated with length of stay. A 5% significance level ($\alpha = 0.05$) was used. Decisions were based on p-values together with effect sizes, because with nearly 50,000 records even trivial differences become statistically significant.

Factor	Test(s) used	Effect size
Gender, 30-day readmission (2 groups)	Independent samples t-test, Welch's t-test, Mann–Whitney U	Cohen's d, rank-biserial r
Age group, diagnosis, admission type, ward, payment type (3+ groups)	One-way ANOVA, Kruskal–Wallis test; Holm-adjusted post-hoc Mann–Whitney tests	η^2 (eta-squared), ε^2 (epsilon-squared)
Age, previous admissions, comorbidity count (numerical)	Pearson's r and Spearman's ρ with 95% confidence intervals	r, ρ, simple-regression slope
All factors together	Multiple linear regression on ln(LOS), F-test, HC3 robust standard errors	R^2, adjusted R^2, Cohen's f^2 (unique contribution)

4. Correlation Analysis

Pearson's correlation measures linear association and Spearman's rank correlation measures monotonic association; both are reported because length of stay is skewed. A correlation matrix of the four numerical variables was also examined to check for overlap between predictors.

5. Multiple Linear Regression

Because LOS is right-skewed, the regression used the natural logarithm of LOS as the dependent variable. Coefficients are converted to percentage changes in LOS using $(e^b - 1) \times 100$. Categorical factors were entered as dummy variables with reference groups (malaria, elective admission, medical ward, female, not readmitted, NHIS). Heteroscedasticity-consistent (HC3) standard errors were used, multicollinearity was checked with variance inflation factors (VIF), and the unique contribution of each factor was measured by the drop in R^2 when it was removed.

6. Model Evaluation

The patients were randomly split into a training set (80%, 38,415 patients) and a test set (20%, 9,604 patients). The model was fitted on the training set and evaluated on the unseen test set using R^2, root mean squared error (RMSE) and mean absolute error (MAE) in days. Residual plots were used to check the model assumptions.

7. Effect Size Interpretation

Cohen's conventions were used: d of 0.2, 0.5 and 0.8 are small, medium and large; η^2 and ε^2 of 0.01, 0.06 and 0.14 are small, medium and large; correlations of 0.1, 0.3 and 0.5 are weak, moderate and strong.

Skills & Tools

Software

Python (pandas, NumPy, SciPy, Matplotlib); Microsoft Excel / CSV (data file format).

Functions Used

Function	Function	Function
read_csv()	str.strip()	drop_duplicates()
map() / replace()	to_numeric()	to_datetime()
fillna()	pd.cut()	groupby()
ttest_ind()	mannwhitneyu()	f_oneway()
kruskal()	pearsonr()	spearmanr()
np.linalg.inv()	np.log()	probplot()

Statistical Techniques

- Descriptive statistics, data visualisation and effect sizes (Cohen's d, η^2, ε^2, rank-biserial r, Cohen's f^2)
- Independent samples t-test, Welch's t-test, Mann–Whitney U test, one-way ANOVA and Kruskal–Wallis test with Holm-adjusted post-hoc comparisons
- Pearson and Spearman correlation analysis
- Multiple linear regression with robust standard errors and train–test evaluation

Applied & Analytical Skills

- Data cleaning, standardisation and missing-data handling
- Healthcare data analysis, hypothesis testing, statistical reporting and data visualisation

Dataset Overview

Before examining individual factors, the composition of the 48,019-patient cleaned dataset was reviewed. Figures 1 to 4 show the main variables and Table 4 summarises the numerical variables. Percentages are rounded and may not add to exactly 100%.

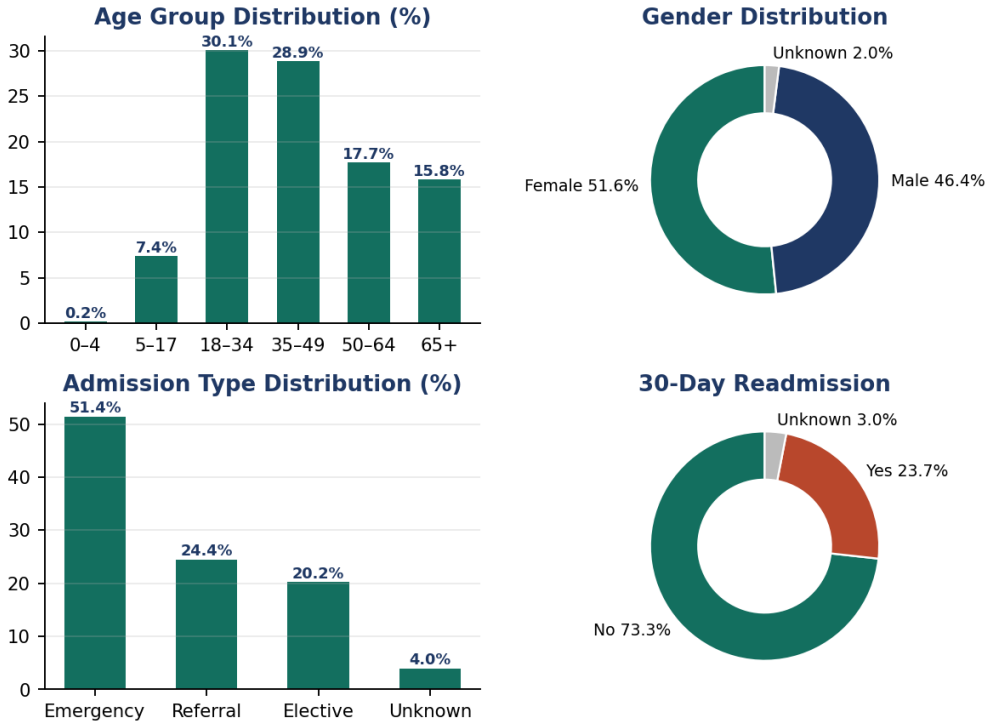


Figure 1. Dataset overview: age group, gender, admission type and 30-day readmission.

Most patients were young to middle-aged adults: 30.1% were aged 18–34 and 28.9% were aged 35–49, while 15.8% were aged 65 and above. Females made up 51.6% and males 46.4% of patients. Emergency admissions were the largest group (51.4%), followed by referrals (24.4%) and elective admissions (20.2%). 23.7% of patients were readmitted within 30 days.

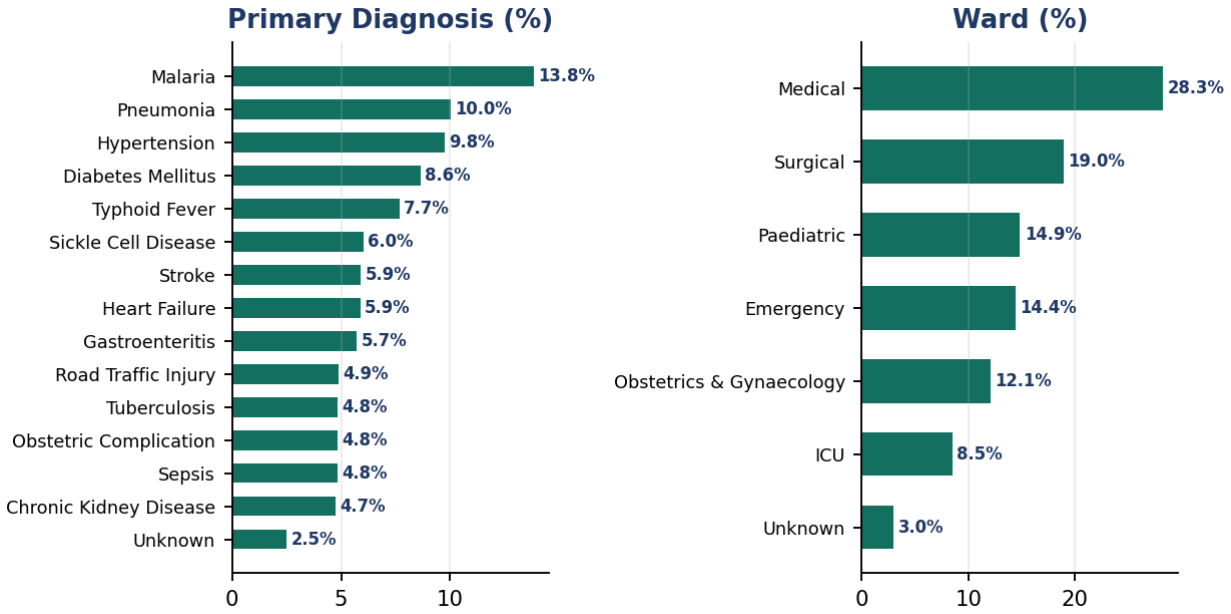


Figure 2. Dataset overview: primary diagnosis and ward.

Malaria was the most common diagnosis (13.8%), followed by pneumonia (10.0%) and hypertension (9.8%). The medical ward received the largest share of admissions (28.3%), followed by the surgical ward (19.0%).

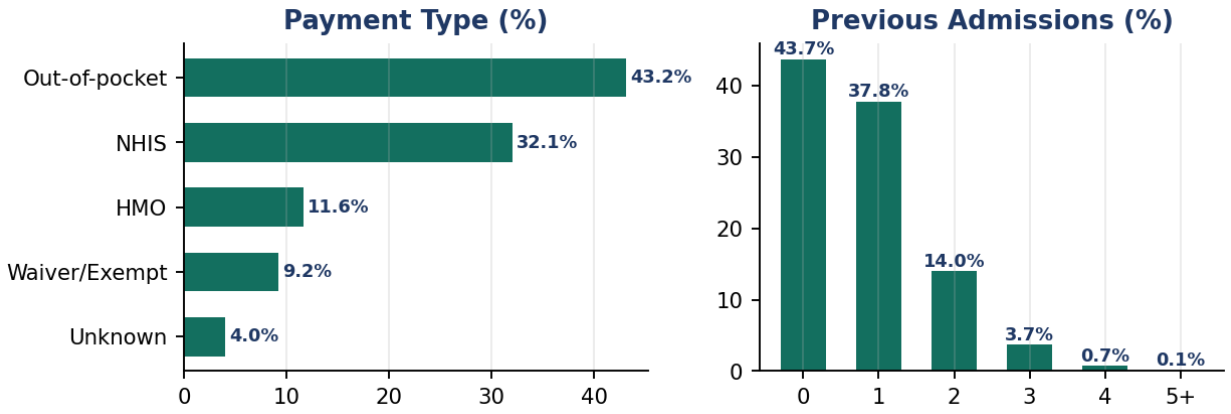


Figure 3. Dataset overview: payment type and number of previous admissions.

Nearly half of the patients paid out of pocket (43.2%), while 32.1% were covered by the National Health Insurance Scheme (NHIS). 4.0% of admission types, 3.0% of wards and 4.0% of payment types were labelled "Unknown" because they were missing in the raw file.

Distribution of length of stay

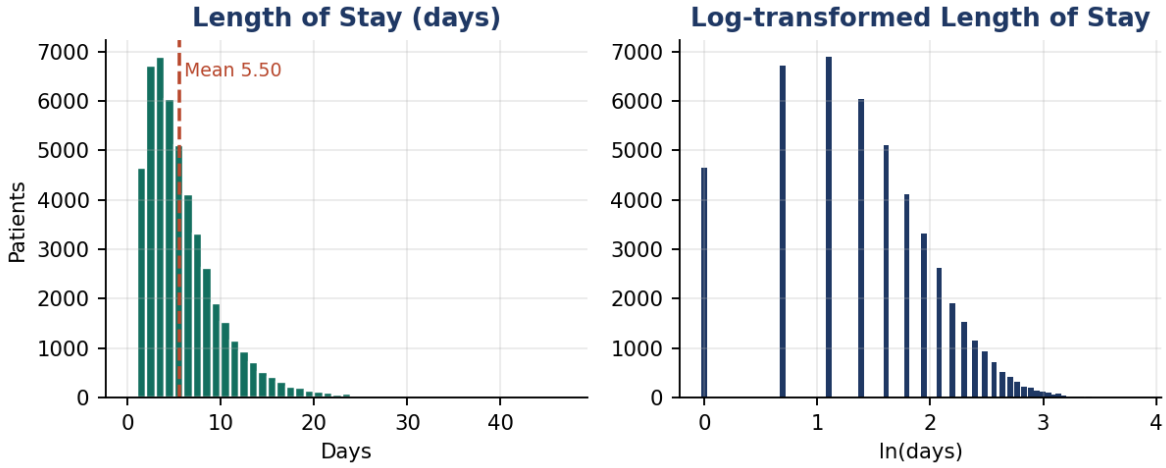


Figure 4. Distribution of length of stay (left) and of log-transformed length of stay (right).

Table 4. Descriptive statistics of the numerical variables (after cleaning).

Variable	Mean	Median	Std	Min	Q1	Q3	Max	Skew.	Kurt.
Age (years)	43.48	40	20.62	0	28	56	98	0.68	0.03
Previous Admissions	0.80	1	0.88	0	0	1	7	1.11	1.29
Comorbidity Count	1.93	2	1.39	0	1	3	8	0.79	0.79
Length of Stay (days)	5.50	4	4.11	1	3	7	46	1.84	5.69

- **Length of stay:** right-skewed (skewness 1.84, kurtosis 5.69). Most stays are short (median 4 days, middle 50% between 3 and 7 days), but a small number of long stays (up to 46 days) pull the mean up to 5.50 days. This is why medians, non-parametric tests and a log transformation are used.
- **Age:** mean and median are close, but the distribution is right-skewed with a large group of young adults. Missing and impossible ages were replaced by the median (40 years).

- **Previous admissions and comorbidity count:** small whole numbers that are mildly right-skewed, with no extreme values after cleaning.

Results

Each section below describes one factor, tests the null hypothesis that the factor is not associated with length of stay, and reports the effect size. Differences are described as associations, not causes.

Section 1: Age vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
0–4	73	5.70	5	3.85
5–17	3,556	4.37	3	3.47
18–34	14,438	4.90	4	3.64
35–49	13,868	5.38	4	3.95
50–64	8,500	5.93	5	4.25
65+	7,584	6.93	6	4.86
All patients	**48,019**	**5.50**	**4**	**4.11**

Average length of stay increased with age, from 4.37 days in children aged 5–17 years to 6.93 days in patients aged 65 and above. (The 0–4 group has only 73 patients, so its average of 5.70 days is less reliable.)

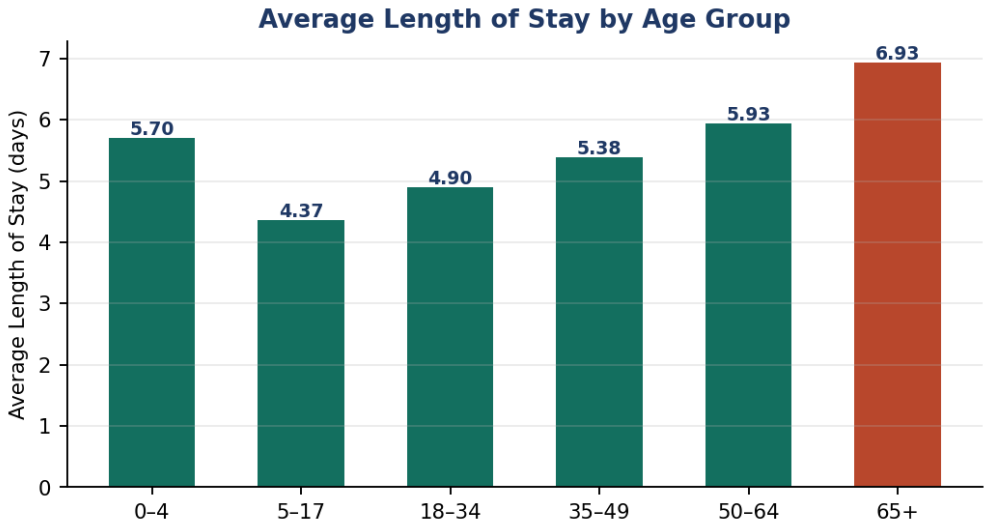


Figure 5. Average length of stay by age group.

Correlation Analysis

Pearson's r: 0.191 (95% CI 0.183 to 0.200), p < 0.001 (weak positive correlation)

Spearman's ρ: 0.188 (95% CI 0.179 to 0.196), p < 0.001

Linear trend: +0.038 days of stay per additional unit (simple regression slope)

Age and length of stay showed a weak positive correlation: each additional 10 years of age was associated with about 0.38 more days in hospital. The scatter plot below shows the trend and the wide spread of individual stays.

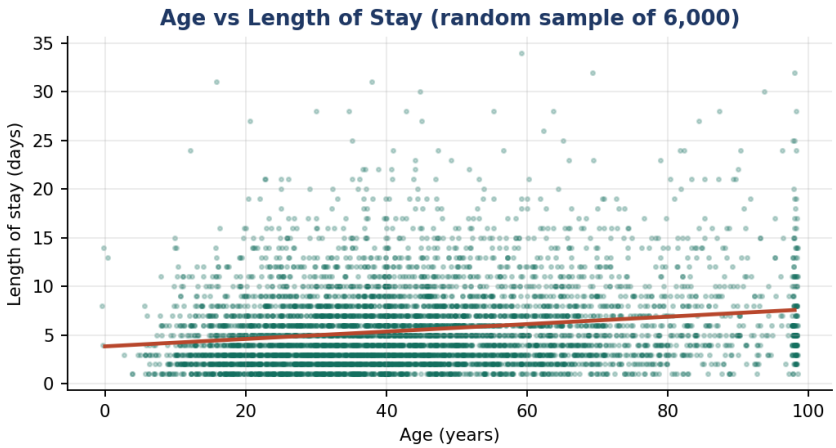


Figure 6. Age against length of stay (random sample of 6,000 patients) with the fitted trend line.

Comparison of Age Groups

One-way ANOVA: $F(5, 48{,}013) = 332.44$, $p < 0.001$, $\eta^2 = 0.0335$ (small effect)

Kruskal–Wallis test: $H = 1{,}674.03$, $df = 5$, $p < 0.001$, $\varepsilon^2 = 0.0348$ (small effect)

Since $p < 0.001$, the null hypothesis was rejected: length of stay differs between age groups. The effect size ($\eta^2 \approx 0.03$) is small, meaning that age group explains about 3% of the variation in length of stay.

Note: 957 patients had a missing or impossible age and were assigned the median age (40 years), so they fall in the 35–49 group.

> **Conclusion:** Age was significantly and positively associated with length of stay. Older patients stayed longer, particularly those aged 65 and above, but the association was weak to small, so age alone does not explain length of stay. This is an association, not proof of causation.

Section 2: Gender vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
Female	24,784	5.50	4	4.10
Male	22,266	5.52	4	4.13
Unknown	969	5.39	4	4.10
All patients	**48,019**	**5.50**	**4**	**4.11**

Average length of stay was almost identical in female and male patients (5.50 and 5.52 days). Patients with unknown gender (n = 969) were excluded from the formal test.

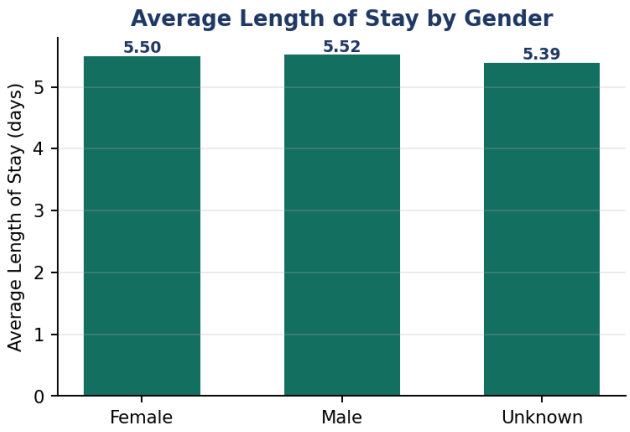


Figure 7. Average length of stay by gender.

Independent Two-Sample t-Test and Mann–Whitney U Test

Mean difference: +0.02 days (Male − Female)

Independent samples t-test: t(47,048) = 0.53, p 0.5949

Welch's t-test: t = 0.53, p 0.5951

Mann–Whitney U test: U = 276,395,709, p 0.7452, rank-biserial r = 0.002

Cohen's d: 0.005 (negligible effect)

Since p = 0.5949 is greater than 0.05, we failed to reject the null hypothesis. There is insufficient evidence that length of stay differs by gender, and Cohen's d shows that any difference is negligible. Welch's t-test and the Mann–Whitney U test gave the same conclusion.

> **Conclusion:** Gender was not significantly associated with length of stay (t = 0.53, p = 0.5949). Male and female patients can be planned for in the same way.

Section 3: Primary Diagnosis vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
Tuberculosis	2,321	7.98	7	5.25
Sepsis	2,320	7.43	6	5.15
Stroke	2,828	7.27	6	4.79
Chronic Kidney Disease	2,274	6.93	6	4.80
Heart Failure	2,822	6.81	6	4.55
Road Traffic Injury	2,339	6.21	5	4.37
Unknown	1,198	5.55	4	4.23
Sickle Cell Disease	2,886	5.43	4	3.89
Diabetes Mellitus	4,148	5.20	4	3.63
Pneumonia	4,810	5.11	4	3.63
Obstetric Complication	2,321	4.90	4	3.61
Hypertension	4,686	4.81	4	3.48
Typhoid Fever	3,680	4.51	4	3.25
Malaria	6,636	4.29	3	3.15
Gastroenteritis	2,750	4.02	3	3.01
All patients	**48,019**	**5.50**	**4**	**4.11**

Tuberculosis had the longest average stay (7.98 days), followed by sepsis (7.43 days), stroke (7.27 days) and chronic kidney disease (6.93 days). Gastroenteritis had the shortest (4.02 days). The gap between the longest and shortest diagnosis was approximately 3.96 days.

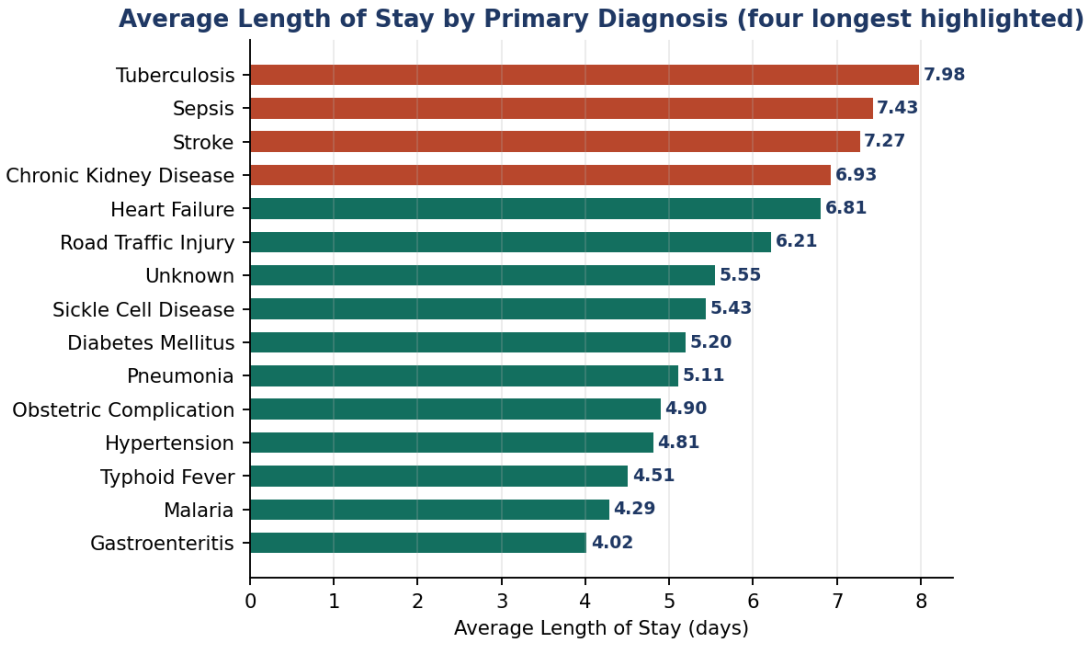


Figure 8. Average length of stay by primary diagnosis (four longest highlighted).

One-Way ANOVA and Kruskal–Wallis Test

One-way ANOVA: F(14, 48,004) = 304.24, $p < 0.001$, $\eta^2 = 0.0815$ (medium effect)

Kruskal–Wallis test: H = 3,876.74, df = 14, $p < 0.001$, $\varepsilon^2 = 0.0805$ (medium effect)

Since $p < 0.001$, the null hypothesis was rejected: length of stay differs between diagnoses. The effect size ($\eta^2 = 0.081$) is medium (0.06–0.14), which makes diagnosis the strongest single categorical factor in this dataset. Each diagnosis was then compared with all other diagnoses combined (Holm-adjusted Mann–Whitney tests). Stays were significantly longer than average for Tuberculosis, Sepsis, Stroke, Chronic Kidney Disease, Heart Failure, Road Traffic Injury. Stays were significantly shorter than average for Diabetes Mellitus, Pneumonia, Obstetric Complication, Hypertension, Typhoid Fever, Malaria, Gastroenteritis. Sickle Cell Disease did not differ significantly from the rest.

> **Conclusion:** Primary diagnosis was significantly associated with length of stay. Tuberculosis, sepsis, stroke, chronic kidney disease and heart failure were linked to the longest stays. These findings can help identify diagnosis groups for care-pathway review, but they do not establish that a diagnosis causes a longer stay on its own, since age, comorbidity and ward also differ between diagnoses.

Section 4: Admission Type vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
Emergency	24,696	5.85	5	4.26
Referral	11,719	5.15	4	3.92
Elective	9,697	5.08	4	3.89
Unknown	1,907	5.39	4	3.97
All patients	**48,019**	**5.50**	**4**	**4.11**

Emergency admissions had the longest average stay (5.85 days), compared with 5.15 days for referrals and 5.08 days for elective admissions.

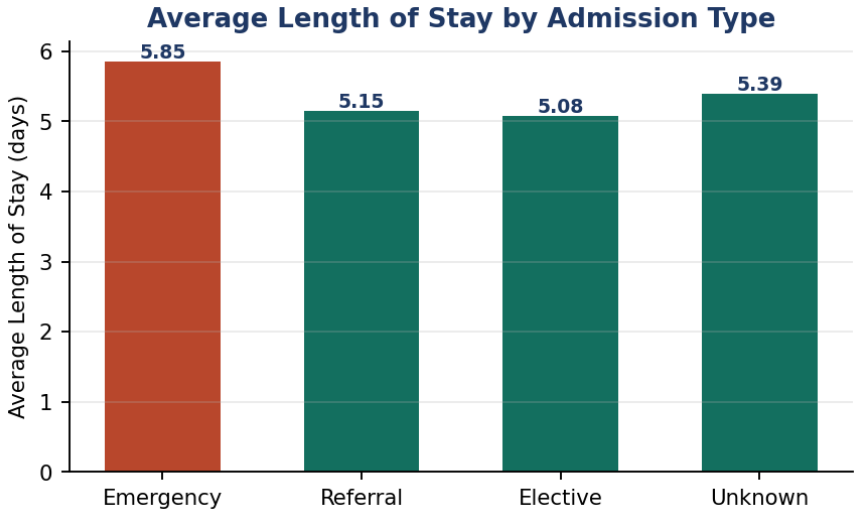


Figure 9. Average length of stay by admission type.

One-Way ANOVA and Kruskal–Wallis Test

One-way ANOVA: F(3, 48,015) = 122.98, $p < 0.001$, $\eta^2 = 0.0076$ (negligible effect)

Kruskal–Wallis test: H = 436.06, df = 3, $p < 0.001$, $\varepsilon^2 = 0.0090$ (negligible effect)

Post-hoc pairwise comparisons (Mann–Whitney U tests, Holm-adjusted)

Comparison	Mean difference (days)	Cohen's d	Holm-adjusted p
Emergency vs Elective	+0.77	0.186	< 0.001
Emergency vs Referral	+0.70	0.168	< 0.001
Referral vs Elective	+0.07	0.019	0.1314

Emergency patients stayed significantly longer than both referral and elective patients (small effect, d ≈ 0.17–0.19). Referral and elective patients did not differ in a meaningful way.

> **Conclusion:** Admission type was significantly associated with length of stay. Emergency admissions stayed about 0.7–0.8 days longer than other admissions, but the overall effect was very small ($\eta^2 < 0.01$). Emergency patients may arrive more acutely ill, so this is not proof that emergency admission itself prolongs the stay.

Section 5: Ward vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
ICU	4,067	7.74	7	5.45
Unknown	1,440	5.38	4	4.05
Medical	13,575	5.32	4	3.95
Paediatric	7,134	5.30	4	3.86
Surgical	9,100	5.29	4	3.93
Obstetrics & Gynaecology	5,790	5.29	4	3.88
Emergency	6,913	5.24	4	3.79
All patients	**48,019**	**5.50**	**4**	**4.11**

ICU patients had by far the longest average stay (7.74 days). The other wards were all clustered between 5.24 and 5.38 days.

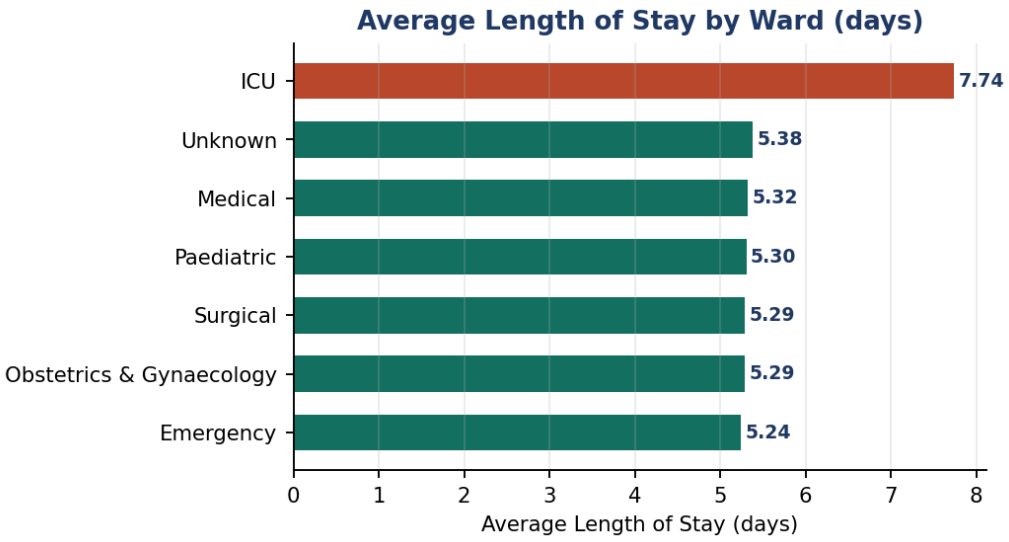


Figure 10. Average length of stay by ward (ICU highlighted).

One-Way ANOVA and Kruskal–Wallis Test

One-way ANOVA: $F(6, 48{,}012) = 225.30$, $p < 0.001$, $\eta^2 = 0.0274$ (small effect)

Kruskal–Wallis test: $H = 962.48$, $df = 6$, $p < 0.001$, $\varepsilon^2 = 0.0199$ (small effect)

Post-hoc comparisons with the medical ward (Holm-adjusted)

Ward vs Medical ward	Mean difference (days)	Cohen's d	Holm-adjusted p
ICU	+2.42	0.557	< 0.001
Unknown	+0.05	0.014	1.0000
Paediatric	-0.02	-0.005	1.0000
Surgical	-0.03	-0.009	1.0000
Obstetrics & Gynaecology	-0.04	-0.009	1.0000
Emergency	-0.08	-0.021	1.0000

Only the ICU differed significantly from the medical ward (about 2.4 days longer, Cohen's d ≈ 0.56, a medium effect). All other wards were statistically similar to the medical ward.

> **Conclusion:** Ward was significantly associated with length of stay, but the association is almost entirely explained by the ICU. Critically ill patients are admitted to the ICU, so the longer stay reflects severity of illness rather than the ward itself.

Section 6: Previous Admissions vs. Length of Stay

Descriptive Analysis

Previous admissions	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
0	20,979	5.19	4	3.92
1	18,135	5.58	5	4.12
2	6,702	5.97	5	4.32
3	1,788	6.40	5	4.74
4+	415	6.76	5	5.09
All patients	**48,019**	**5.50**	**4**	**4.11**

Average length of stay rose steadily with the number of previous admissions, from 5.19 days in patients with none to 6.76 days in patients with four or more (the "4+" group combines all higher values).

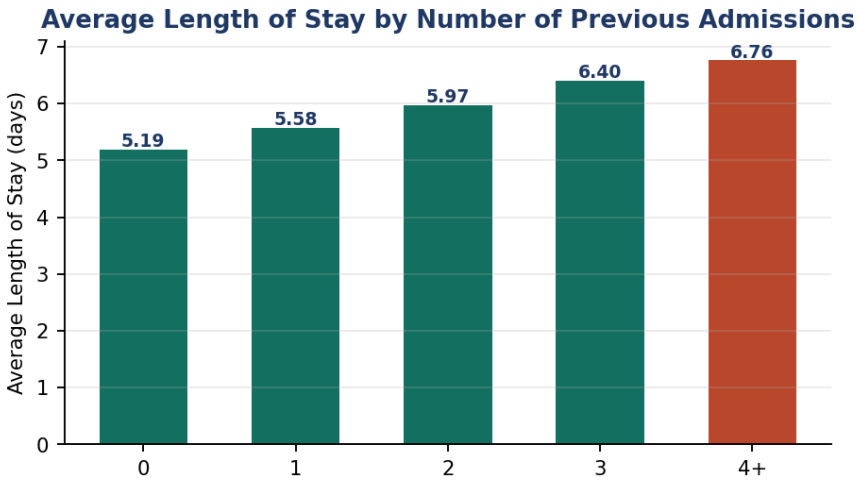


Figure 11. Average length of stay by number of previous admissions.

Correlation Analysis

Pearson's r: 0.084 (95% CI 0.075 to 0.093), $p < 0.001$ (very weak positive correlation)

Spearman's ρ: 0.082 (95% CI 0.073 to 0.091), $p < 0.001$

Linear trend: +0.391 days of stay per additional unit (simple regression slope)

One-way ANOVA: $F(4, 48{,}014) = 84.59$, $p < 0.001$, $\eta^2 = 0.0070$ (negligible effect)

Kruskal–Wallis test: $H = 345.18$, $df = 4$, $p < 0.001$, $\varepsilon^2 = 0.0071$ (negligible effect)

The correlation is statistically significant but very weak ($\rho \approx 0.08$), and previous admissions explain less than 1% of the variation in length of stay ($\eta^2 \approx 0.007$).

> **Conclusion:** More previous admissions were significantly associated with longer stays, with each extra admission adding about 0.4 days on average. The effect is small, but it is consistent and may help flag patients who need closer follow-up.

Section 7: Comorbidity Count vs. Length of Stay

Descriptive Analysis

Comorbidities	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
0	6,942	4.34	3	3.36
1	12,805	4.95	4	3.65
2	14,560	5.52	5	4.01
3	7,599	6.17	5	4.41
4	3,799	6.72	6	4.58
5+	2,314	7.81	7	5.35
All patients	**48,019**	**5.50**	**4**	**4.11**

Length of stay increased steadily with each additional coexisting condition, from 4.34 days with no comorbidity to 7.81 days with five or more (the "5+" group combines all higher values).

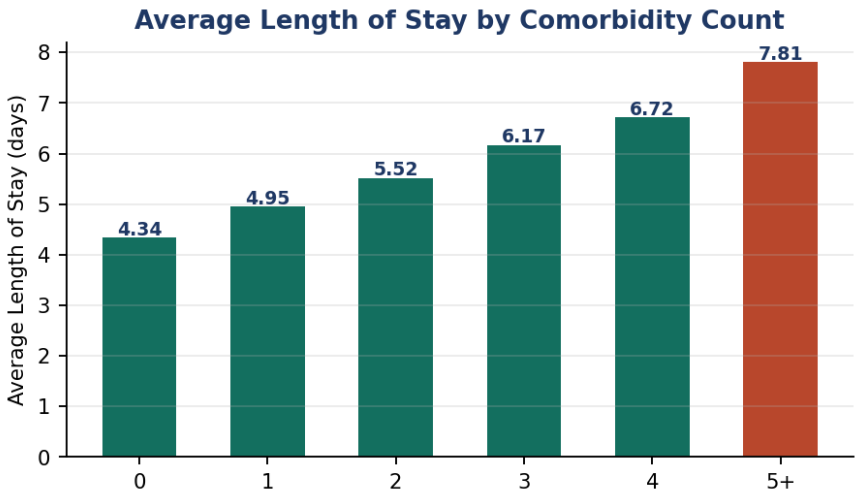


Figure 12. Average length of stay by comorbidity count.

Correlation Analysis

Pearson's r: 0.208 (95% CI 0.200 to 0.217), $p < 0.001$ (weak positive correlation)

Spearman's ρ: 0.205 (95% CI 0.196 to 0.213), $p < 0.001$

Linear trend: +0.618 days of stay per additional unit (simple regression slope)

One-way ANOVA: $F(5, 48{,}013) = 427.68$, $p < 0.001$, $\eta^2 = 0.0426$ (small effect)

Kruskal–Wallis test: $H = 2{,}105.71$, $df = 5$, $p < 0.001$, $\varepsilon^2 = 0.0438$ (small effect)

Comorbidity count had the strongest correlation with length of stay of the three numerical variables ($\rho \approx 0.20$, a weak relationship). Patients with five or more conditions stayed on average 3.5 days longer than patients with none.

> **Conclusion:** Comorbidity count was significantly and positively associated with length of stay, with a small effect ($\eta^2 = 0.043$), the largest of the three numerical factors. Patients with several coexisting conditions are the clearest candidates for early care planning.

Section 8: 30-Day Readmission vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
No	35,177	5.00	4	3.52
Yes	11,387	7.05	6	5.26
Unknown	1,455	5.65	5	4.05
All patients	**48,019**	**5.50**	**4**	**4.11**

Patients who were readmitted within 30 days had a longer length of stay (7.05 days) than those who were not (5.00 days), a difference of 2.05 days.

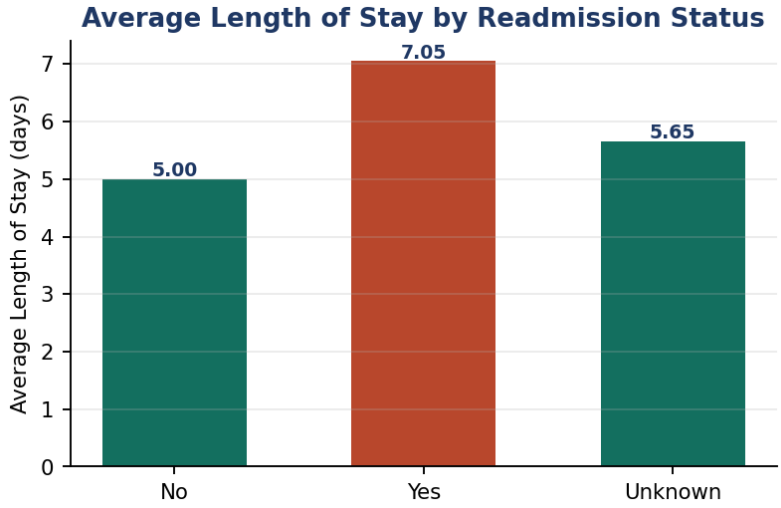


Figure 13. Average length of stay by 30-day readmission status.

Independent Two-Sample t-Test and Mann–Whitney U Test

Mean difference: +2.05 days (Readmitted − Not readmitted)

Independent samples t-test: t(46,562) = 47.44, p < 0.001

Welch's t-test: t = 38.96, p < 0.001

Mann–Whitney U test: U = 248,104,787, p < 0.001, rank-biserial r = 0.239

Cohen's d: 0.511 (medium effect)

Since p < 0.001, the null hypothesis was rejected. Cohen's d = 0.51 is a medium effect, the largest effect size among the two-group comparisons in this report. Welch's t-test and the Mann–Whitney U test agree. Patients with unknown readmission status (n = 1,455) were excluded from the test.

> **Conclusion:** Patients who were later readmitted had stayed about 2 days longer on their first admission. Readmission happens after discharge, so this should be read as an association: longer stays may mark sicker patients, or patients discharged before they were fully ready, but this dataset cannot tell which.

Section 9: Payment Type vs. Length of Stay

Descriptive Analysis

Group	Patients (n)	Mean LOS (days)	Median (days)	Std. deviation
Unknown	1,912	5.59	5	4.12
Waiver/Exempt	4,400	5.52	4	4.25
Out-of-pocket	20,724	5.52	4	4.14
NHIS	15,395	5.50	4	4.07
HMO	5,588	5.42	4	3.99
All patients	**48,019**	**5.50**	**4**	**4.11**

Average length of stay was almost the same for every payment group, ranging from 5.42 days (HMO) to 5.59 days (unknown).

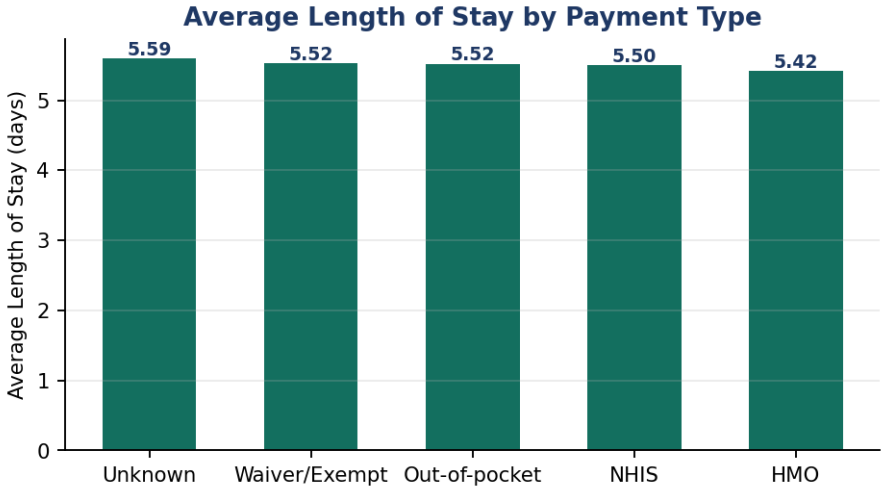


Figure 14. Average length of stay by payment type.

One-Way ANOVA and Kruskal–Wallis Test

One-way ANOVA: F(4, 48,014) = 0.94, p 0.4410, η^2 = 0.0001 (negligible effect)

Kruskal–Wallis test: H = 2.08, df = 4, p 0.7202, ε^2 = 0.0000 (negligible effect)

Since p = 0.4410 (ANOVA) and p = 0.7202 (Kruskal–Wallis) are both greater than 0.05, we failed to reject the null hypothesis. The effect size is effectively zero.

> **Conclusion:** Payment type (NHIS, out-of-pocket, HMO or waiver) was not associated with length of stay in this dataset. There is no sign that the way a patient pays influences how long they are kept in hospital.

Section 10: Correlation Matrix of Numerical Variables

A correlation matrix was built for the four numerical variables to see how they relate to length of stay and to one another. Overlap between predictors (multicollinearity) can make regression coefficients unstable, so this check was done before the regression.

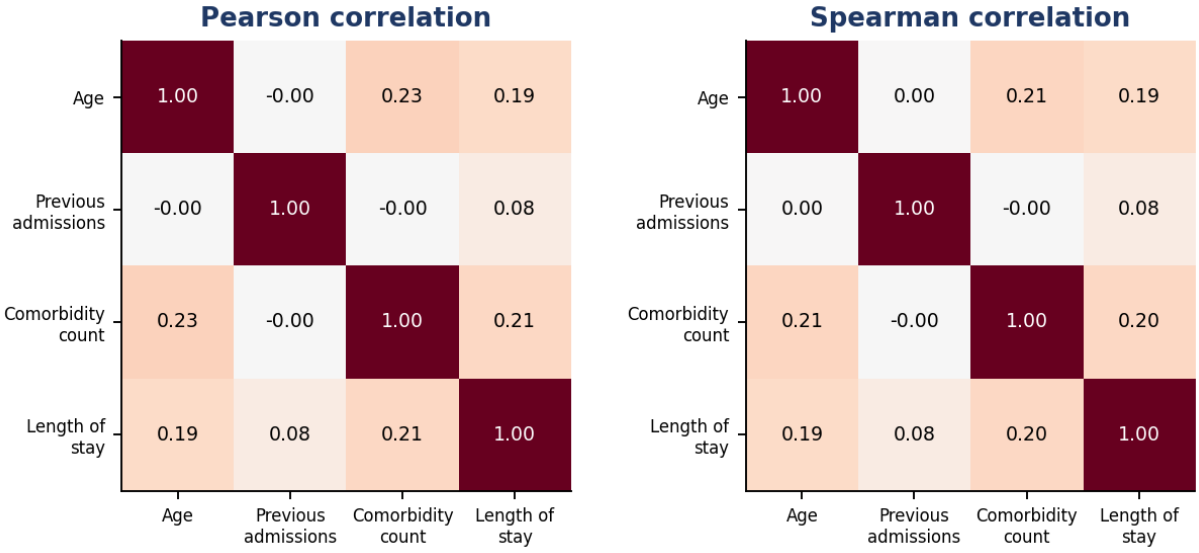


Figure 15. Pearson and Spearman correlation matrices of age, previous admissions, comorbidity count and length of stay.

Table 10. Spearman correlation matrix.

Variable	Age	Previous admissions	Comorbidity count	Length of stay
Age	1.000	0.001	0.214	0.188
Previous admissions	0.001	1.000	-0.002	0.082
Comorbidity count	0.214	-0.002	1.000	0.205
Length of stay	0.188	0.082	0.205	1.000

All three factors correlate positively with length of stay (ρ = 0.20 for comorbidity count, 0.19 for age and 0.08 for previous admissions). Age and comorbidity count are weakly correlated with each other (ρ = 0.21), which is expected because older patients tend to have more conditions. Previous admissions are unrelated to age or comorbidity count in this dataset. Because the predictors overlap only weakly, they carry mostly separate information about length of stay.

> **Conclusion:** Comorbidity count and age were the most strongly correlated with length of stay, though all correlations were weak. No pair of predictors was strongly correlated, so multicollinearity is not a concern for the regression.

Section 11: Multiple Linear Regression for Length of Stay

The sections above examined one factor at a time. Because factors overlap (older patients have more comorbidities, ICU patients have particular diagnoses), a multiple linear regression was fitted to estimate the association of each factor with length of stay after accounting for all the others.

- **Dependent variable:** natural logarithm of length of stay
- **Independent variables:** age (per 10 years), previous admissions, comorbidity count, diagnosis, admission type, ward, gender, 30-day readmission and payment type
- **Reference groups:** malaria, elective admission, medical ward, female, not readmitted, NHIS
- **Sample size:** 48,019 patients

Table 11. Selected regression results (full table in Appendix A).

Variable	Change in LOS (%)	95% CI	p-value
Age (per 10 years)	+5.4%	+5.1% to +5.7%	< 0.001
Previous admissions (per admission)	+6.7%	+6.0% to +7.5%	< 0.001
Comorbidity count (per condition)	+8.9%	+8.5% to +9.4%	< 0.001
Tuberculosis (vs malaria)	+87.8%	+81.9% to +93.9%	< 0.001
Stroke (vs malaria)	+72.2%	+67.1% to +77.4%	< 0.001
Sepsis (vs malaria)	+71.5%	+66.0% to +77.3%	< 0.001
Chronic kidney disease (vs malaria)	+64.0%	+58.7% to +69.3%	< 0.001
Heart failure (vs malaria)	+62.5%	+57.8% to +67.3%	< 0.001
ICU (vs medical ward)	+44.1%	+40.6% to +47.6%	< 0.001
Emergency admission (vs elective)	+16.4%	+14.6% to +18.3%	< 0.001
Readmitted within 30 days (vs not)	+22.4%	+20.7% to +24.3%	< 0.001
Male (vs female)	+0.5%	-0.8% to +1.7%	0.4618
Out-of-pocket (vs NHIS)	+0.4%	-1.0% to +1.8%	0.5532

Interpretation of Results

- **Comorbidity count:** each additional coexisting condition was associated with about 8.9% longer stay, holding the other factors constant.
- **Previous admissions:** each additional previous admission was associated with about 6.7% longer stay.
- **Age:** every 10 additional years of age were associated with about 5.4% longer stay.
- **Diagnosis:** compared with malaria, tuberculosis (88%), stroke (72%), sepsis (72%), chronic kidney disease (64%) and heart failure (63%) were associated with 60–90% longer stays.
- **Ward and admission type:** ICU patients stayed about 44% longer than medical-ward patients, and emergency admissions about 16% longer than elective admissions.
- **Readmission:** patients readmitted within 30 days had stays about 22% longer.
- **Gender and payment type:** no significant association after adjustment, which confirms the single-factor results.

Coefficients are percentage changes in the geometric mean of length of stay per one-unit increase (or compared with the reference group). They show association, not causation.

Model Strength

R^2: 0.1965 (about 19.7% of the variation in ln(LOS))

Adjusted R^2: 0.1959

Overall F-test: $F(34, 47{,}984) = 345.2$, $p < 0.001$

Residual standard error: 0.669 (log scale)

Maximum VIF: 1.72 (values below 5 indicate no multicollinearity problem)

The model is statistically far better than a model with no predictors, but an R^2 of about 0.20 shows that the recorded factors explain only about one-fifth of the variation in length of stay. Clinical severity, procedures performed, complications, delays in tests and discharge, and social factors were not recorded in this dataset and are likely to explain much of the rest.

Table 12. Unique contribution of each factor to the regression model.

Factor	Unique contribution to R^2 (points)	Cohen's f^2	F-test p-value
Diagnosis	7.68	0.0956	< 0.001
Comorbidity count	2.37	0.0295	< 0.001
Age	1.96	0.0245	< 0.001
Ward	1.93	0.0240	< 0.001
30-day readmission	1.28	0.0159	< 0.001
Admission type	0.93	0.0116	< 0.001
Previous admissions	0.59	0.0073	< 0.001
Payment type	0.00	0.0001	0.6402
Gender	0.00	0.0000	0.6913

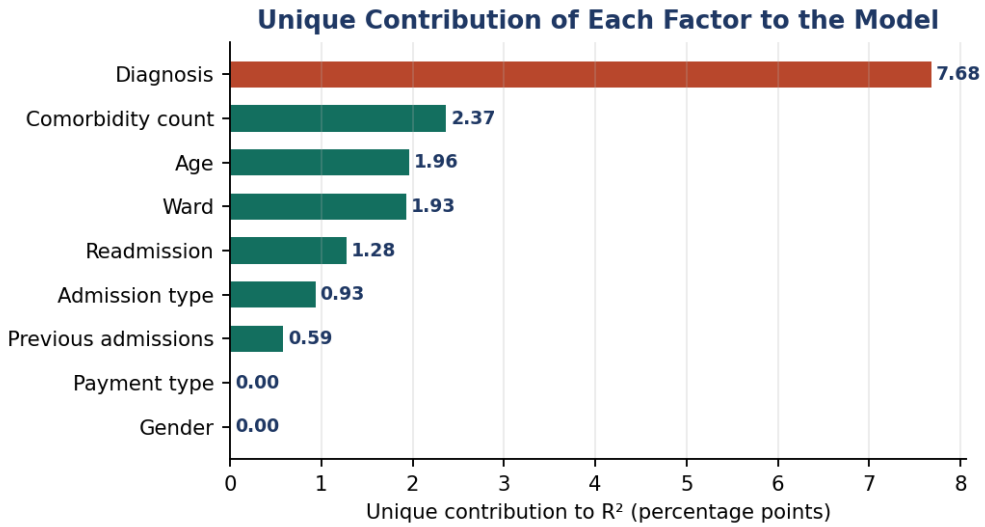


Figure 16. Unique contribution of each factor to the explained variation in log length of stay.

Diagnosis made by far the largest unique contribution (about 7.7 percentage points of R^2), followed by comorbidity count, age, ward, readmission, admission type and previous admissions. Gender and payment type contributed nothing. All contributions are small in absolute terms (Cohen's f^2 of 0.10 or below for every factor), which is why no single factor can predict length of stay well.

Model Evaluation on Unseen Data

Measure	Result	Meaning
R^2 (training set, ln LOS)	0.196	Fit on the 80% of patients used to build the model
R^2 (test set, ln LOS)	0.199	Fit on 20% of patients the model had not seen
RMSE (test set, ln LOS)	0.671	Typical prediction error on the log scale
MAE (test set, days)	2.68 days	Average error when predicting LOS in days
RMSE (test set, days)	3.66 days	Prediction error, giving more weight to big misses
R^2 (test set, days)	0.201	Share of variation in LOS in days that is predicted

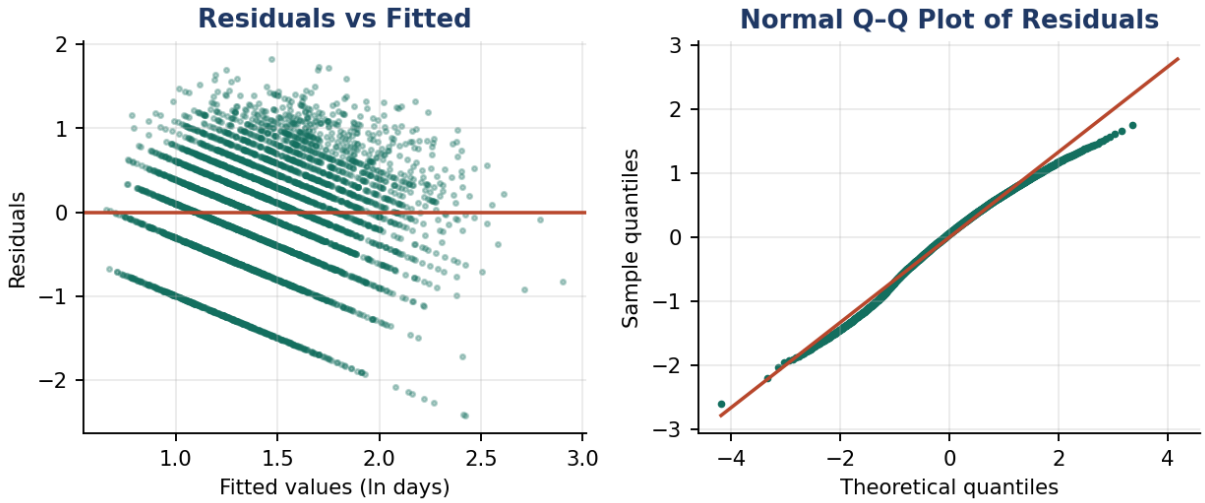


Figure 17. Residual diagnostics: residuals against fitted values (left) and normal Q–Q plot (right).

The test-set R^2 (0.199) is almost the same as the training R^2 (0.196), so the model is not overfitted. On average its predictions were about 2.7 days away from the true length of stay, compared with a mean stay of 5.5 days, so the model is useful for understanding patterns but is not accurate enough to predict the stay of an individual patient.

The diagonal bands in the residual plot are expected because length of stay is recorded in whole days. The residuals are roughly symmetric (skewness -0.37) and the Q–Q plot follows the line closely in the middle with slightly light tails, so the log transformation worked reasonably well, and robust standard errors were used for added protection.

Conclusion: After adjusting for all other factors, primary diagnosis, comorbidity count, age, ICU admission, 30-day readmission, emergency admission and previous admissions were each significantly associated with length of stay, while gender, payment type and the non-ICU wards were not. Diagnosis was the most important factor. The model explains about 20% of the variation in LOS and generalised well to unseen data, but because the data are synthetic, the results describe patterns within this dataset and are not clinical evidence.

Limitations

- **Synthetic data.** The dataset was generated with Claude AI and contains no real patients. The findings describe patterns within this dataset and are not clinical evidence for any hospital.
- **Association, not causation.** Even in the regression, the results show associations. Important drivers such as illness severity, procedures, complications and delays in discharge were not recorded.
- **Statistical versus practical significance.** With about 48,000 patients, small differences become statistically significant. Most effect sizes were small (largest $\eta^2 = 0.081$ for diagnosis; largest Cohen's d = 0.51 for readmission), so effect sizes should be read together with p-values.
- **Imputation.** Missing ages, previous admissions and comorbidity counts were replaced with medians. This is simple and transparent, but it slightly reduces variation, and all 957 imputed ages fall in the 35–49 group.
- **Removed records.** 1,231 records with no usable length of stay and 750 duplicates were removed. If these records differ systematically from the others, the results could be affected.
- **"Unknown" categories.** Missing diagnosis, ward, admission type and payment type were labelled "Unknown" rather than estimated. They were kept in the regression as their own groups.
- **Readmission as a factor.** Readmission happens after the index stay is over. It was analysed as an associated characteristic, not as a cause of the length of stay.
- **Right-skewed outcome.** Length of stay is skewed and recorded in whole days. A log transformation, non-parametric tests and robust standard errors were used, but other approaches (such as count or Gamma models) could be compared.
- **Multiple testing.** Many tests were run. Holm adjustment was used for post-hoc comparisons, and the main conclusions rest on very small p-values and effect sizes rather than on borderline results.
- **No external validation.** The model was tested on a hold-out sample from the same dataset, not on data from another hospital.

Recommendations

Because the data are synthetic, the recommendations below are suggestions for further investigation rather than established clinical guidance.

1. **Build diagnosis-specific care pathways** for tuberculosis, sepsis, stroke, chronic kidney disease and heart failure, which had the longest stays, with target discharge dates and early identification of delays.
2. **Flag patients at admission who are likely to stay longer**: older patients, patients with three or more comorbidities and patients with several previous admissions, so that discharge planning and social-support assessments start early.
3. **Review ICU step-down and discharge criteria**, because ICU stays were about 2.4 days longer than medical-ward stays; reducing avoidable ICU days frees up the hospital's most limited beds.
4. **Streamline the emergency admission pathway**, since emergency patients stayed longer than elective and referred patients, by speeding up early investigations and specialist review.
5. **Check discharge readiness in patients with long stays**, because patients who were later readmitted had stayed about 2 days longer; a standard discharge checklist and early follow-up may help.
6. **Do not use gender or payment type to plan bed capacity.** Neither was associated with length of stay, so NHIS, HMO and out-of-pocket patients can be planned for in the same way.
7. **Improve data quality at the point of entry**, using drop-down lists for diagnosis (ideally ICD-10 codes), ward, gender and admission type, with mandatory age, length-of-stay and date fields. This would have prevented most of the problems found during cleaning.

8 **Build a routine dashboard** that tracks average and median length of stay by diagnosis, ward and age group each month.

Next Steps

This project focused on single-factor tests and one multiple linear regression model. Future work could extend the analysis by:

1 Comparing the log-linear model with Gamma, Poisson or negative binomial regression models designed for positive, skewed outcomes.

2 Adding interaction terms (for example, age by diagnosis, or ICU by comorbidity) to find out whether the effect of one factor depends on another.

3 Applying machine learning models such as random forest or gradient boosting to predict length of stay and comparing them with the regression model.

4 Using survival (time-to-discharge) analysis to study how quickly different patient groups are discharged.

5 Building an interactive Power BI or Excel dashboard for hospital administrators.

6 Validating the findings with real, anonymised data from Nigerian teaching hospitals, including severity scores, procedures, ICD-10 diagnoses and reasons for delayed discharge.

Appendix A: Full Regression Results

Dependent variable: natural logarithm of length of stay (days). Percentage changes are calculated as $(e^b - 1) \times 100$ and show how much longer (+) or shorter (−) the stay is compared with the reference group, or per one-unit increase for numerical variables. 95% confidence intervals use HC3 robust standard errors. The intercept is shown as the estimated length of stay in days for the reference patient (aged 0 years with no admissions or comorbidities, malaria, elective, medical ward, female, not readmitted, NHIS).

Variable	Coefficient (b)	Change in LOS	95% CI	p-value
Intercept (reference patient)	0.6033	1.83 days	1.78 to 1.88 days	< 0.001
Age (per 10 years)	0.0523	+5.4%	+5.1% to +5.7%	< 0.001
Previous admissions	0.0651	+6.7%	+6.0% to +7.5%	< 0.001
Comorbidity count	0.0857	+8.9%	+8.5% to +9.4%	< 0.001
Diagnosis: Chronic Kidney Disease	0.4944	+64.0%	+58.7% to +69.3%	< 0.001
Diagnosis: Diabetes Mellitus	0.2004	+22.2%	+19.1% to +25.4%	< 0.001
Diagnosis: Gastroenteritis	-0.0543	-5.3%	-8.0% to -2.5%	< 0.001
Diagnosis: Heart Failure	0.4855	+62.5%	+57.8% to +67.3%	< 0.001
Diagnosis: Hypertension	0.1169	+12.4%	+9.7% to +15.2%	< 0.001
Diagnosis: Obstetric Complication	0.1226	+13.0%	+9.5% to +16.7%	< 0.001
Diagnosis: Pneumonia	0.1829	+20.1%	+17.1% to +23.1%	< 0.001
Diagnosis: Road Traffic Injury	0.3802	+46.3%	+41.7% to +51.0%	< 0.001
Diagnosis: Sepsis	0.5397	+71.5%	+66.0% to +77.3%	< 0.001
Diagnosis: Sickle Cell Disease	0.2498	+28.4%	+24.7% to +32.1%	< 0.001
Diagnosis: Stroke	0.5432	+72.2%	+67.1% to +77.4%	< 0.001
Diagnosis: Tuberculosis	0.6303	+87.8%	+81.9% to +93.9%	< 0.001
Diagnosis: Typhoid Fever	0.0533	+5.5%	+2.7% to +8.3%	< 0.001
Diagnosis: Unknown	0.2273	+25.5%	+20.3% to +31.0%	< 0.001
Admission type: Emergency	0.1521	+16.4%	+14.6% to +18.3%	< 0.001
Admission type: Referral	0.0084	+0.8%	-0.9% to +2.7%	0.3582
Admission type: Unknown	0.0803	+8.4%	+4.9% to +12.0%	< 0.001
Ward: Emergency	-0.0053	-0.5%	-2.4% to +1.4%	0.5871
Ward: ICU	0.3651	+44.1%	+40.6% to +47.6%	< 0.001
Ward: Obstetrics & Gynaecology	-0.0156	-1.6%	-3.5% to +0.5%	0.1328
Ward: Paediatric	-0.0068	-0.7%	-2.5% to +1.2%	0.4834
Ward: Surgical	-0.0129	-1.3%	-3.0% to +0.5%	0.1523
Ward: Unknown	-0.0059	-0.6%	-4.2% to +3.2%	0.7529
Gender: Male	0.0045	+0.5%	-0.8% to +1.7%	0.4618
Gender: Unknown	-0.0075	-0.7%	-4.9% to +3.6%	0.7316
Readmitted 30d: Unknown	0.0891	+9.3%	+5.5% to +13.3%	< 0.001
Readmitted 30d: Yes	0.2025	+22.4%	+20.7% to +24.3%	< 0.001
Payment type: HMO	0.0002	+0.0%	-2.0% to +2.1%	0.9881

Variable	Coefficient (b)	Change in LOS	95% CI	p-value
Payment type: Out-of-pocket	0.0042	+0.4%	-1.0% to +1.8%	0.5532
Payment type: Unknown	0.0234	+2.4%	-0.8% to +5.6%	0.1429
Payment type: Waiver/Exempt	-0.0024	-0.2%	-2.4% to +2.0%	0.8358

Model: n = 48,019, R^2 = 0.1965, adjusted R^2 = 0.1959. Reference groups: malaria, elective, medical ward, female, not readmitted, NHIS.